



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ELECTRODYNAMICS I				
Type of Course	Core in Major				
Semester	IV				
Academic Level	200 – 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	A strong foundation in mathematics, including algebra, trigonometry, and calculus. Additionally, a basic understanding of physics concepts such as electricity, magnetism, and mechanics would be beneficial for grasping the principles covered in the course.				
Course Summary	The course provides a foundational exploration of electromagnetism, encompassing topics like electric fields, magnetic fields and electromagnetic induction. Through simplified explanations, illustrative examples, and conceptual exercises, students gain insight into the behavior and interactions of electric and magnetic fields, preparing them for more advanced studies in physics or related fields at the undergraduate level.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply vector analysis techniques to solve problems in electromagnetics	Application	Conceptual Understanding	Problem-solving assignments, quizzes
CO2	Analyze and calculate electric fields and potentials for various charge distributions	Analysis	Procedural Knowledge	Homework assignments, exams, simulation exercises
CO3	Investigate the behavior of magnetic fields and solve problems involving magnetostatics	Evaluation	Conceptual Understanding	Laboratory reports, group projects, exams
CO4	Utilize electrical measurement instruments to quantify electric and magnetic phenomena	Application	Procedural Knowledge	Laboratory experiments, instrument operation tests, practical assessments
CO5	Demonstrate an understanding of Maxwell's equations and their implications in electromagnetism	Comprehension	Conceptual Understanding	Concept maps, oral presentations, written exams
CO6	Apply theoretical knowledge to analyze and design simple electromagnetic systems	Synthesis	Procedural Knowledge	Design projects, case studies, final projects
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	VECTOR ANALYSIS		12	20
	1	Vector Algebra	2	
	2	Differential Calculus	4	
	3	Integral Calculus	4	

	4	Curvilinear Coordinates	2	
	Sections 1.1.1 – 1.1.4, 1.2.1 – 1.2.7, 1.3.1 – 1.3.6, 1.4.1 – 1.4.2 of chapter 1 of Book 1			
II	ELECTROSTATICS		15	20
	5	The Electric Field	3	
	6	Divergence and Curl of Electrostatic Field	4	
	7	Electric Potential; Electrostatic Boundary Conditions	4	
	8	Work and Energy in Electrostatics	2	
	9	Conductors	2	
	Sections 2.1.1 – 2.1.4, 2.2.1, 2.2.3, 2.2.4, 2.3.1 – 2.3.5, 2.4.1 – 2.4.4, 2.5.1 – 2.5.4 of chapter 2 of Book 1 (section 2.2.2 is excluded)			
III	MAGNETOSTATICS		9	15
	10	The Lorentz Force Law	2	
	11	The Biot – Savart Law	2	
	12	The Divergence and Curl of B (up to the derivation of Eqn. 5.50); Ampere's Law	2	
	13	Magnetic Vector Potential; Magnetostatic Boundary Conditions	3	
	Sections 5.1.1 – 5.1.3, 5.2.1, 5.2.2, 5.3.1 – 5.3.4, 5.4.1, 5.4.2 of chapter 5 of Book 1			
IV	ELECTRICAL MEASUREMENTS		9	15
	14	Kirchoff's laws and Wheatstone's Bridge	1	
	15	Carey Foster Bridge	1	
	16	Potentiometer	1	
	17	Network Analysis: Superposition Theorem	1	
	18	Thevenin's Theorem, Norton's theorem	1	
	19.	Maximum power transfer theorem	1	
	20	Maxwell's Loop Current Method	1	
	21	Torque on a Current loop in a Unifor, Magnetic field	1	
	22	Moving Coil Ballistic Galvanometer	1	

	Sections 6.6 – 6.8, 6.12 – 6.17 of chapter 6, and sections 10.10, 10.11 of chapter 10 of Book 2		
V	PRACTICALS	30	
	Conduct any 6 experiments from the given list (two from experiment 1-4 and four from 5-16) and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.		
	1 Plotting of the 2D functions using Python - Plot the 2D function in Problem 1.12 of Book 1 and find the maximum value and the location of maxima from the plot. - Simulations of section 1.2 of Book 3 can be referred.		
	2 Mapping of 2D vector fields using Python - Map the vector fields in Example 1.4 and 1.5 of Book 1. - Map $\frac{\hat{r}}{r}$ and $\frac{\hat{r}}{r^2}$ - Simulations of section 3.1 of Book 3 can be referred.		
	3 Mapping of electric and magnetic field lines using Python - Plot the field of an electric charge, dipole and magnetic dipole. - Simulations of section 4.1, 4.2 and Appendix D of Book 3 can be referred.		
	4 Simulation of particle trajectory under Lorentz force law using Python - Simulate the trajectory of charged particle moving under Lorentz force law. - Problem 5.66 of Book 1 and Chapter 6 of Book 3 can be referred		
	5 Mapping of the magnetic field lines of a bar magnet. - Fix a paper on a drawing board kept on a table and place the bar magnet at the center along the magnetic meridian. - Using a small compass needle, map the magnetic field lines of the magnet placed with (a) north pole pointing south and (b) north pole pointing north. - Mark the null points (where the horizontal component of Earth's magnetic field, B_h cancels the field due to magnet) along the axial/equatorial line and measure the distance, $2d$, between them. - Calculate the moment of the magnet. (a) $m = \frac{4\pi}{\mu_0} \frac{(d^2 - l^2)^2}{2d} B_h$ (b) $m = \frac{4\pi}{\mu_0} (d^2 + l^2)^{3/2} B_h$		

6	Study the variation of the magnetic field strength of a bar magnet using a smartphone magnetometer. - Using a smartphone magnetometer, measure the strength of the magnetic field of a bar magnet, along the axial and equatorial lines and plot the data. - Magnetometer in the Phyphox app may be used to get the data after locating the approximate position of the magnetometer sensor. https://phyphox.org/wiki/index.php?title=Sensor:_Magnetic_field - Fit the theoretical formulae to the data and obtain magnetic dipole moment. Along the axial line $B = \frac{\mu_0}{4\pi} \frac{2md}{(d^2 - l^2)^2}$ and along the equatorial line $B = \frac{\mu_0}{4\pi} \frac{m}{(d^2 + l^2)^{3/2}}$		
7	Determine the moment of a bar magnet and Bh using a deflection magnetometer and a box type vibration magnetometer. - Determine m/Bh using deflection magnetometer in Tan A position and mBh using box type vibration magnetometer. Hence calculate the moment of the magnet and Bh. - If the same magnet was used, compare the dipole moment with that of experiment 2 and 3.		
8	Circular coil- Verification of Biot Savart's law and determination of Bh. - Move a compass through a platform along the axis of the coil carrying a steady current. Note the deflection of the needle and plot magnetic flux density ($B = B_h \tan \theta$) as a function of distance. - Optional: Smartphone magnetometer may be used to measure the strength of the magnetic field along the axial line and plot the data. https://phyphox.org/experiment/magnetic-field/ - By varying current and (or) distance of the compass box along the axial line of the coil, note the deflection and hence determine the value of Bh.		
9	Reduction factor of TG using potentiometer. - Standardize the given potentiometer using a Daniell cell or any other constant voltage source and use the standardized potentiometer to find the current through the TG. - By observing the deflection in the TG for different currents, calculate the reduction factor. - From the magnetic field at the center of a circular coil, deduce the value Bh.		
10	Verification of Kirchoff's laws / Superposition theorem.		

	- Verify Kirchoff's current law at a junction where a minimum of three branches meet. - Verify Kirchoff's current law for a network with two loops. OR - Verify the superposition theorem for a network with two sources, S1 and S2. - First set particular voltage values in S1 and S2 and note down the ammeter reading. - Set the same voltage in S1 and short circuit S2 and vice versa, note down the ammeter readings and verify the superposition theorem.		
11	Verification of Thevenin's theorem and maximum power transfer theorem Thevenin's theorem - Measure the current through the load resistance of the network. - Estimate the values of R_{TH} and V_{TH}, construct the Thevenin's equivalent circuit and measure the current through load resistance and compare the two results with the theoretical values. Maximum power transfer theorem - Measure the current through load resistance and estimate the power. Plot $R_L - P$ graph and find the R_L corresponding to the maximum power. - Calculate the % of error with the theoretical value.		
12	Determination of resistivity of a thin wire using Carey-Foster's Bridge - Find the resistance per unit length of the bridge wire. - Determine resistance of the thin wire using the bridge, thickness of the wire using screw gauge and hence determine the resistivity.		
13	Calibrate the ammeter using potentiometer - Standardize the potentiometer using a Daniell cell or any other standard voltage source. - Determine the current for at least 8 trials and draw the calibration graph.		
14	Conversion of Galvanometer to voltmeter and calibration using potentiometer - Determine the value of high resistance required to connect in series with the galvanometer so as it can read 0.1V or 0.2V per scale division. - Standardize the potentiometer using a Daniell cell or any other standard voltage source. - Determine the voltage for at least 6 trials and draw the calibration graph.		

15	BG-Determination of the figure of merits for current - Determine the figure of merits for current of the given ballistic galvanometer. - Measure a small current using BG and verify with ammeter.		
16	BG-Comparison of capacitance- Desauty's method Compare the capacitance of two given capacitors by forming De-Sauty bridge.		

Book for Reference:

1. Introduction to Electrodynamics (5th Edn.) by David J Griffiths, Cambridge University Press (Book 1)
2. Electricity and Magnetism (10 Edn.) by R Murugesan, S. Chand and Company (Book 2)
3. Electrodynamics Tutorials with Python Simulations by Taejoon Kouh, Minjoon Kouh -CRC Press 1st Edition (Book 3)
4. Electricity and Magnetism, Berkeley Physics Course Vol.2, by E M Purcell, Mc Graw Hill Edn.
5. Electricity and Magnetism, by D C Pandey, Arihand Prakashan Series
6. Classical Electromagnetism by H C Verma, Bharathi Bhavan Publishers and Distributors
7. The Feynman Lectures on Physics, Vol-2, Pearson Education India
8. NPTEL lectures on Electrodynamics/ Classical Electrodynamics
<https://archive.nptel.ac.in/courses/115/105/115105132/>

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	3	2	3	0	2	3	3	2	3	2	0
CO 2	1	3	2	3	3	0	2	3	3	2	3	2	0
CO 3	3	2	3	1	3	0	3	3	3	2	3	2	0
CO 4	1	3	2	3	2	1	2	3	3	2	3	2	0
CO 5	2	2	3	1	3	0	3	3	3	2	3	2	0
CO 6	3	1	3	3	3	0	3	3	3	2	3	2	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	